

MB 206: Excel for Data Analysis

Course Code: MB 206

Course Name: Excel for Data Analysis

L-T-P-C: 3-0-0-3

Prerequisites: NA

Intended for: IMBA

Mutual Exclusion: NA

Approval: 59th BoA (I-MBA curriculum); course proposal approved in the 60th BoA

Preamble

This 3-credit course is designed to equip IMBA students with essential skills in Microsoft Excel for effective data analysis and business decision-making. Through a hands-on, application-oriented approach, students will learn to manage, analyze, and visualize data using Excel's advanced functions, tools, and add-ins. The course lays a strong foundation for data-driven thinking by integrating analytical techniques with real-world business scenarios, thereby preparing students to make informed decisions in dynamic organizational settings.

Course Modules

- **Introduction:** Elementary spreadsheet concepts and processes, worksheets and workbooks, importing data into Excel, Entering and arranging data, Data validation, Relative and absolute References, Formatting Range of cells, filtering and sorting data. (2)
- **Excel Formulas and Functions:** Index and Match functions, Text functions, IF statements, Count functions, Numeric functions, String functions, Date time functions, Lookup functions, Logical Functions, Sum and average functions, Rank functions, Indirect functions, and Functions for Specific Applications. (8)
- **Statistical Analysis:** Summarizing Data with Descriptive Statistics and Database Statistical Functions, Confidence Interval, and Generating random numbers, random variables: Binomial, Poisson, exponential and normal random variables. (4)
- **Visualizing Data:** Approaching the Task of Visualization, Preparing Data for Robust Graphics, Working with Standard Charts, Explore chart styles, Customize charts, Combine chart types, 3D Surface plot, contour plots, Geographic visualization tool. (4)
- **PivotTables:** Understanding PivotTables, Anatomy of a PivotTable, Creating a PivotTable, Calculations in PivotTables, Updating a PivotTable, Using data models and analysing disparate data sources, Power Pivot, Power View and 3D Maps. (6)
- **Modeling Relationships and Trends in Data:** Correlation to summarise relationships, Relationship between correlation and R-squared, Simple Linear Regression: Finding the Best-Fitting Regression Line, Regression Models for Prediction, Regression as Analysis of Variance, Testing Hypotheses for Regression Coefficients, Confidence Intervals for Regression Coefficients, Residual Analysis, and Regression Assumptions, Checking Assumptions, Multiple Linear Regression, Building Good Regression Models, Correlation and Multicollinearity, Practical Issues in Trendline and Regression Modeling, Regression with Categorical Independent Variables, Categorical Variables with More Than Two Levels. (6)

- **Optimization with Solver:** Understanding Solver Adin, Simplex LP engine for linear optimization problems, GRG Nonlinear engine for nonlinear optimization problems and Evolutionary Solver engine to tackle nonsmooth optimization problems. (4)
- **What-if-Analysis:** Create a data table, Define a scenario, Analyse data with scenarios, Sensitivity analysis with data tables, Scenario Manager for sensitivity analysis, Goal Seek for data analysis. (4)
- **Automating Tasks with Macros:** What is macro? Recording a macro, running a macro, adding and editing a macro. (4)

Laboratory/Practical/Tutorial Modules

NA

Text Books

NA

References

1. Wayne Winston – *Microsoft Excel Data Analysis and Business Modeling* – PHI Learning Pvt. Ltd.
2. Paul McFedries, *Excel Data Analysis*, John Wiley and Sons, 2013.
3. Bill Jelen and Michael Alexander, *Microsoft Excel 2019 Pivot Table Data Crunching*, Microsoft Press, 2019.
4. Paul Mcfedries, *Excel Data Analysis: Your Visual Blueprint for Analyzing Data, Charts and Pivot Tables*, Wiley (2013).